

Cartwheel Homework 1

Implementation and evaluation report

5	1	12	32
required tools	additional tool	recorded conversations	final focused checks

Outcome

The Cartwheel support agent now has all five required tools, one evidence-driven additional tool, a targeted prompt correction, and twelve assessed conversations. The implementation is committed locally at **b839451**.

This report records what was built, how it was tested, what the live conversations revealed, and what remains before final Homework 1 submission. It intentionally excludes credentials and generated database contents.

Repository: **/Users/shcrockett/cartwheel-homeworks**
Model used for live conversations: **gpt-5.6-luna** with Fast service requested
Report date: **September 16, 2026**

1. What we implemented

Homework 1 required deterministic tools behind the model. The tools enforce data access and state rules in code, while the prompt guides decisions such as whether to retrieve policy, refuse, or escalate.

Tool	Implemented behavior	Important boundary
get_policy	Exact, case-sensitive policy lookup with title, audience, and body.	Unknown IDs return structured not_found results; no invented policy.
search_products	Token matching over title and description; optional store and price filters; price-first ordering.	Empty queries and nonpositive ceilings are rejected; valid no-match searches succeed with an empty list.
list_my_orders	Recent orders for the authenticated shopper or merchant store.	Support cannot use this broad listing tool and must look up specific orders.
cancel_order	Cancels an authorized order only while its status is placed.	Kill switch, existence, permission, and shipment state are checked before mutation.
find_order	Fuzzy product-name search over the caller's complete authorized order scope.	Matching happens before the five-result limit, preserving newest-first order.

Additional tool: get_my_store

A live merchant conversation exposed a missing capability: the model knew only `store_id=2` and could not reliably map phrases such as 'our store' to Juniper Home Goods. The merchant-only `get_my_store` tool now derives identity from authenticated context and returns the exact store policy ID. It is unavailable to shoppers and support staff.

Why this mattered

Before the tool, the agent selected Meridian Cycles and gave the wrong 21-day return window. After the tool, it retrieved Juniper Home Goods and correctly cited the 14-day override.

2. Conversations and coverage

Twelve real model conversations were recorded in **hw1-session.jsonl**. Each record contains the authenticated role, request, observed tool calls and results, final response, expected behavior, requirement, student assessment, and problem source.

Coverage	Result	Evidence
Authorized shopper lookup	Pass	Order 4127 accurately reported as delivered.
Out-of-window refund	Fail	Internal eligibility flag leaked; unsolicited escalation opened.
Above-threshold refund	Pass with note	\$240 refund queued; \$100 threshold citation omitted.
Merchant cross-store access	Pass	Permission denied without disclosing order details.
Store override - before	Fail	Wrong store policy selected from generic retrieval.
Store override - after	Pass	get_my_store led to the correct 14-day Juniper policy.
Out-of-scope weather	Pass	Concise refusal with no tool calls.
Account change - before	Fail	No policy lookup and no required human escalation.
Support order access	Pass	Support could access any order as specified.
Constrained product search	Pass with note	Four products correctly filtered and sorted.
Eligible cancellation	Pass	Order 10 changed from placed to cancelled after tool success.
Account change - after	Pass	Policies cited and ticket 151 created with 24-hour SLA.

Role coverage: **8 shopper**, **3 merchant**, and **1 support** conversation. Assessment totals: **9 passes** and **3 failures**. The failures were retained because they provide the evidence needed for tool and prompt improvements.

3. What the live evaluation revealed

Authorization held

The cross-store merchant request was denied by the tool layer, and the final response did not reveal inaccessible order data.

Correct data can still produce a poor answer

For order 3980, the tool correctly returned `refund_eligible=false`, but the model exposed the internal flag, omitted the customer-facing return-window explanation, and opened an unsolicited escalation.

Retrieval needs authenticated context

Generic policy search could not infer which store 'our store' meant. The resulting wrong answer was a capability gap, not a database or permission bug.

Citations are easy to miss

The above-threshold refund correctly entered human review, but the response repeated the \$100 policy threshold without citing cw-refunds.

Output usefulness differs from contract correctness

The product search satisfied its contract, but duplicate Portable Mug titles would be easier to distinguish if the response included product IDs. Product deeplinks are outside the current specification.

Concise responses were inconsistent

Several correct answers included extra details. This did not violate the current specification, but it is a useful future response-quality dimension.

4. Evidence-backed prompt improvement

The strongest prompt failure involved an email-address change. The specification says account changes always go to a human, but the original prompt did not state that rule explicitly.

	Before revision	After revision
Prompt version	d108f6949e5e	f4cf9d1978a4
Policy lookup	None	cw-account-security and cw-escalations
Escalation	None; user was told to contact support	Ticket 151 created by escalate_to_human
Final behavior	Did not satisfy ESC-2	Reported human follow-up within 24 hours

Minimal prompt change

Account changes of any kind always require human handling. Search for the applicable policy, call `escalate_to_human`, and report the created ticket; do not merely tell the user to contact support.

The same user, request, model, and starting database state were used before and after the change. The rerun retrieved the correct policies, created the ticket, cited the policy IDs, reported the SLA, and did not claim the email itself had been changed.

5. Validation and deliverables

Check	Observed result
Focused HW1 contract suite	8 passed with --runxfail
Tool, authorization, and eligibility regressions	24 passed
Prompt and CLI-related offline checks after revision	45 passed
Conversation record validation	12 valid JSONL objects; required fields present
Formatting check	git diff --check passed
Database state	Reset to deterministic dev seed after write conversations
Local commit	b839451 - Complete Homework 1 support agent

Committed submission files

- agent/tools.py - five required tools and get_my_store
- agent/agent.py - tool registration, Fast service passthrough, and the evidence-backed ESC-2 prompt instruction
- hw1-session.jsonl - twelve observed and assessed conversations

Remaining HW1 action

Record one continuous video of no more than five minutes. Show an authorized request, a permission denial, the order 4455 refund queue, the account-change before/after prompt test, one focused test, and the 12-record conversation count.

Transition to Homework 2

Homework 2 adds authenticated HTTP sessions and OpenTelemetry tracing. The first implementation milestone is to add authenticated identity and permission-decision attributes to active tool spans in **observability/instrument.py**. Docker and Langfuse are required later for live trace inspection, but not for the initial offline implementation and authentication tests.